

Use of intelligent systems in reservoir characterization via synthetic magnetic resonance logs

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Abstract

Magnetic Resonance Imaging (MRI) logs are well logs that use nuclear magnetic resonance to accurately measure free fluid, irreducible water (MBVI), and effective porosity (MPHI). Permeability is then calculated using a mathematical function that incorporates these measured properties. This paper describes the methodology developed to generate synthetic MRI logs using data obtained by conventional well logs such as SP, Gamma Ray, Caliper, and Resistivity. The synthetically generated logs are named Virtual Magnetic Imaging Logs or “VMRI” logs for short. © 2001 Elsevier Science B.V. All rights reserved.

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1. Summary

This methodology incorporates artificial neural networks as its main tool. Virtual Magnetic Resonance Imaging (MRI) logs for irreducible water saturation (MBVI) and effective porosity (MPHI) as well as permeability (MPERM) were generated for four wells. These wells are located in East Texas, Gulf of Mexico, Utah, and New Mexico. The results are quite encouraging.

It is shown that MPHI, MBVI, and MPERM logs can be generated with a high degree of accuracy. For

each case, 30% of the data were used to develop the neural model. The model was then tested on the remaining 70% of the data for verification. The models provide VMRI logs with approximately 80% to 97% accuracy using data not employed during model development.

This methodology does not supersede the need for performing MRI logging in a field. It is designed to supplement the process by reducing the cost of using MRI logging on an entire field. The natural application of this process is in fields that have conventional logs for all of the wells but MRI logs for only a few wells. To generate the Virtual MRI logs for every well in a field, data from wells that have both conventional and MRI logs are used in the model development and verification. The model is then applied to the other wells in the field to generate the virtual MRI logs for these wells.

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Using this process, the operator can log a few strategically chosen wells using physical MRI tools and produce virtual MRI logs for the entire field. This will allow the development of an accurate representation of effective porosity, free fluid, irreducible water saturation, and permeability for the entire field.

2. Introduction

John M. Austin and Tom L. Faulkner (Austin and Faulkner, 1993) published a superb paper in August 1993 in “The American Oil and Gas Reporter” providing some valuable information about the MRI logging technique and its benefits to low resistivity reservoirs.

The MRI log measures effective porosity—total porosity minus the clay bound porosity—as well as irreducible water saturation. The irreducible water saturation is the combinations of clay bound water and water held due to the surface tension of the matrix material. The difference between effective porosity (MPHI) and the irreducible water saturation (MBVI) is called the free fluid index. This is the producible fluid in the reservoir. This demonstrates how valuable MRI log is to low resistivity reservoirs. In many low resistivity reservoirs, matrix irreducible water rather than producible water may cause a drop in resistivity. While producible water can seriously hamper production and make the pay quite unattractive, the same cannot be said for the irreducible water. Therefore, a reservoir that seems to be a poor candidate for further development—looking only at the conventional logs—may prove to be an attractive prospect once MRI log is employed.

As it will be shown in this paper, MRI logs may provide information that results in an increase in the recoverable reserves. This takes place simply by including the portions of the pay zone into the recoverable reserve calculations that were excluded during the analysis using only the conventional well logs.

General background about neural networks has been published in numerous papers and will not be repeated in this paper. An example of such publications is included in the references (Mohaghegh et al., 1995, 1996a,b, 1997).

3. Methodology

Four wells from different locations in the United States are used to demonstrate the development of Virtual MRI logs. These wells are from East Texas, New Mexico, Utah, and Gulf of Mexico. These wells are from different formations and since the virtual MRI methodology is a formation specific process, the authors had no options but to test the methodology using single wells. In this paper, the authors will use a part of the pay zone to develop the model and verify it using the remainder of the pay. The ideal way to show the actual potential of this methodology is to use several wells from the same formation. The prerequisite is that both conventional and MRI logs for the wells should be available. In such a situation, a few of the wells would be used to develop the model and the remaining wells would be used for verification purposes. The authors are in the process of locating data to demonstrate the accuracy of the virtual MRI methodology in these situations.

For each well in this study, Gamma Ray, Spontaneous Potential, Caliper, and Resistivity logs were available. These logs were digitized with a resolution of 6 in. for the entire pay zone. Thirty percent of the data were chosen randomly for the model development and the remaining 70% of the pay were used for verification. In all four cases, the model was able to generate synthetic MRI logs with correlation coefficients of up to 0.97 for data that was not used during the model development process.

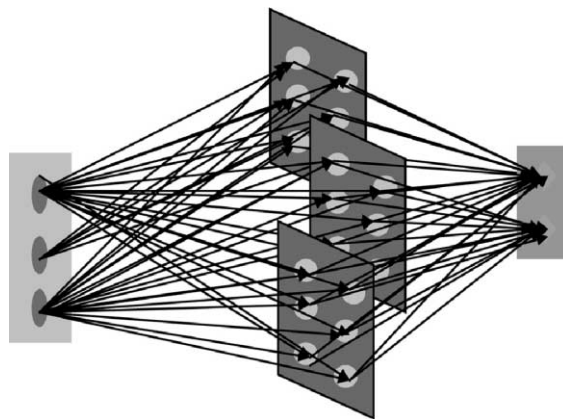


Fig. 1. Schematic diagram of the neural networks used in this study.

The model development process was implemented using a fully connected, five-layer neural network architecture with different activation functions in each layer. These layers included one input layer, three hidden layers and one output layer. Each of the hidden layers has been designed to detect distinct features of the model. A schematic diagram of the network architecture is shown in Fig. 1.

Please note that in this figure all of the neurons and/or connections are shown. The purpose of the figure is to show the general architecture of the

network used for this study. A supervised gradient descent backpropagation of error method was used to train the neural networks. The input layer has six neurons representing Depth, Gamma Ray, SP, Caliper, Medium and Deep Resistivity. Each hidden layer included five neurons. Upon completion of the training process, each neural network contained six weight matrices. Three of the weight matrices had 30 elements while the remaining matrices each had five. In most cases, acceptable generalization was achieved in less than 500 visits to the entire training data.

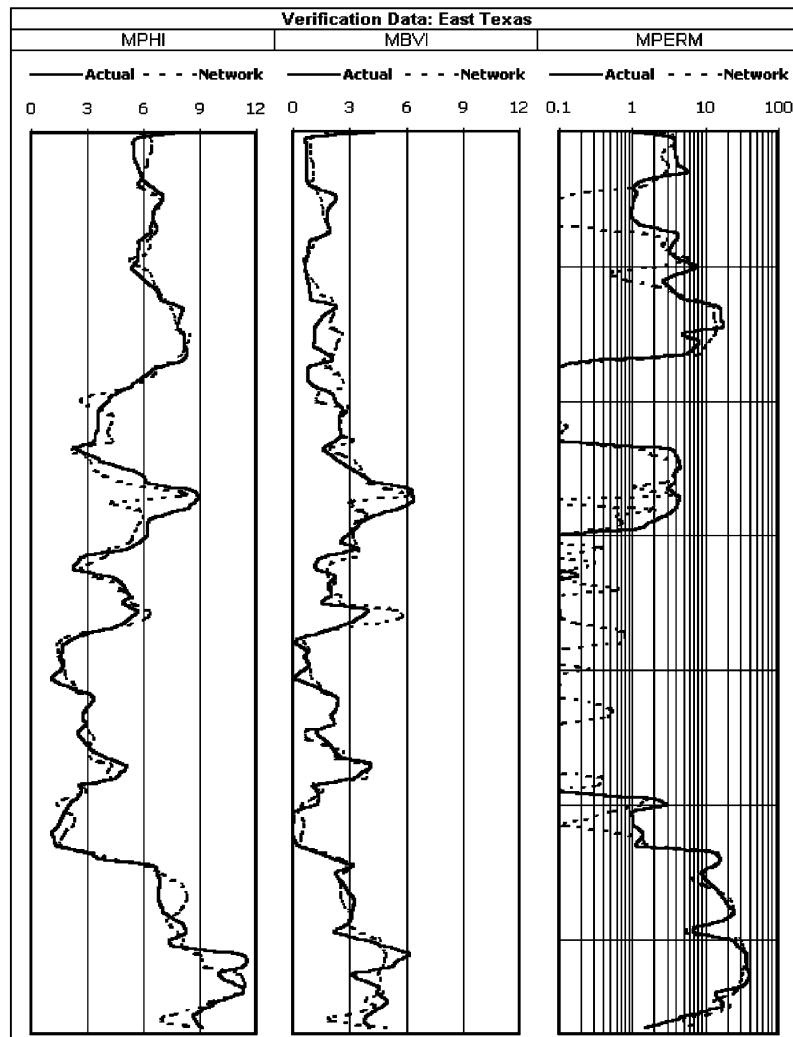


Fig. 2. Virtual and actual MRI logs for the verification data set for the well in East Texas.

Once the network was trained, it was used to generate the virtual MPHI, MBVI and MPERM logs. The MPHI and MBVI logs were then used to estimate the reserves. In one case—the well in New Mexico—that was a tight reservoir, the resolution of the permeability data made it impossible to train an adequate network.

It should be noted that it might be more effective not to use a neural network to replicate the MPERM

log. Since this log is derived from MPHI and MBVI, it would be better to calculate the virtual MPERM log from the virtual MPHI and MBVI.

4. Results and discussion

Figs. 2 and 3 show the actual and virtual MPHI, MBVI, and MPERM logs for the well in East Texas.

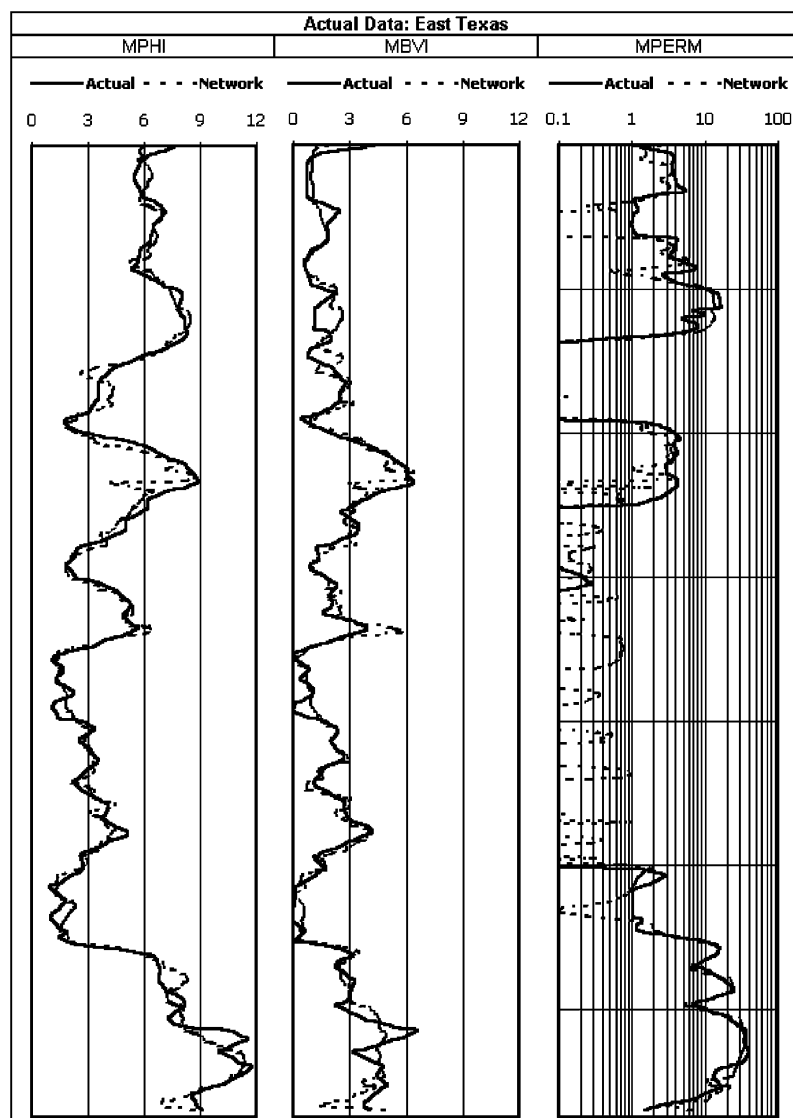


Fig. 3. Virtual and actual MRI logs for the entire pay zone for the well in East Texas.

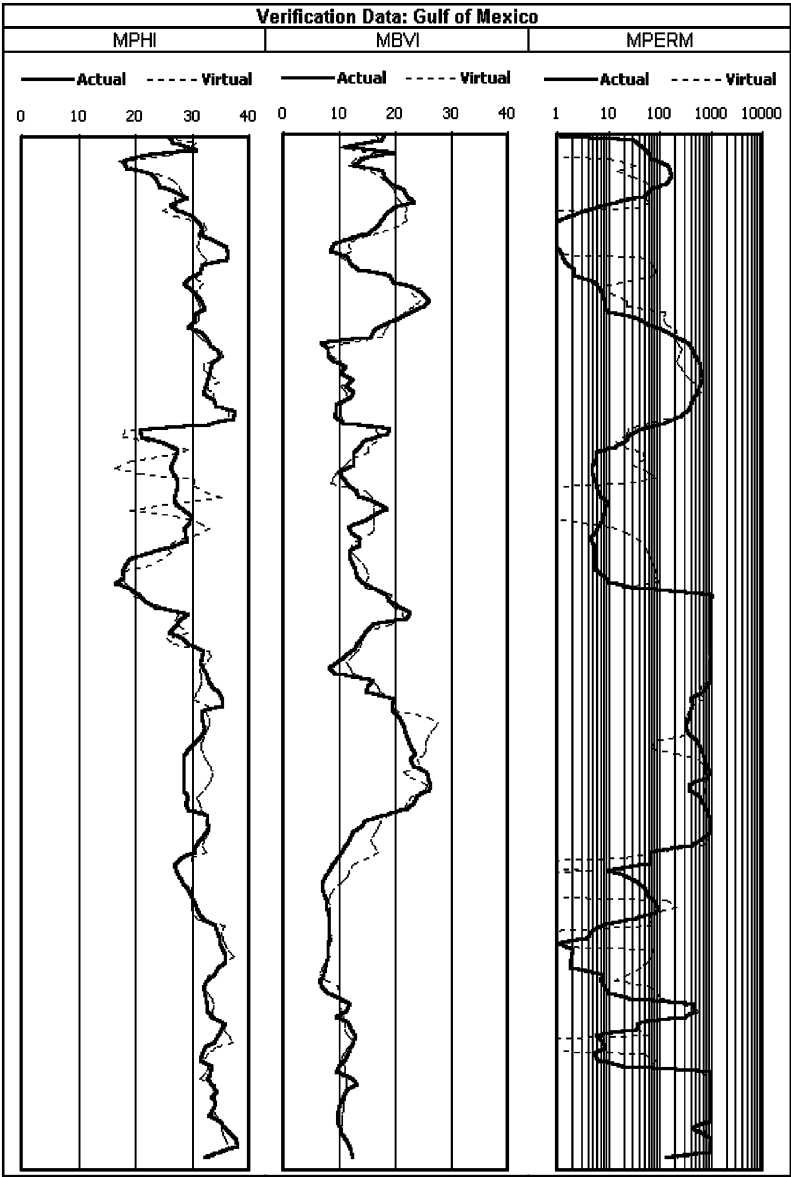


Fig. 4. Virtual and actual MRI logs for the verification data set for the well in Gulf of Mexico.

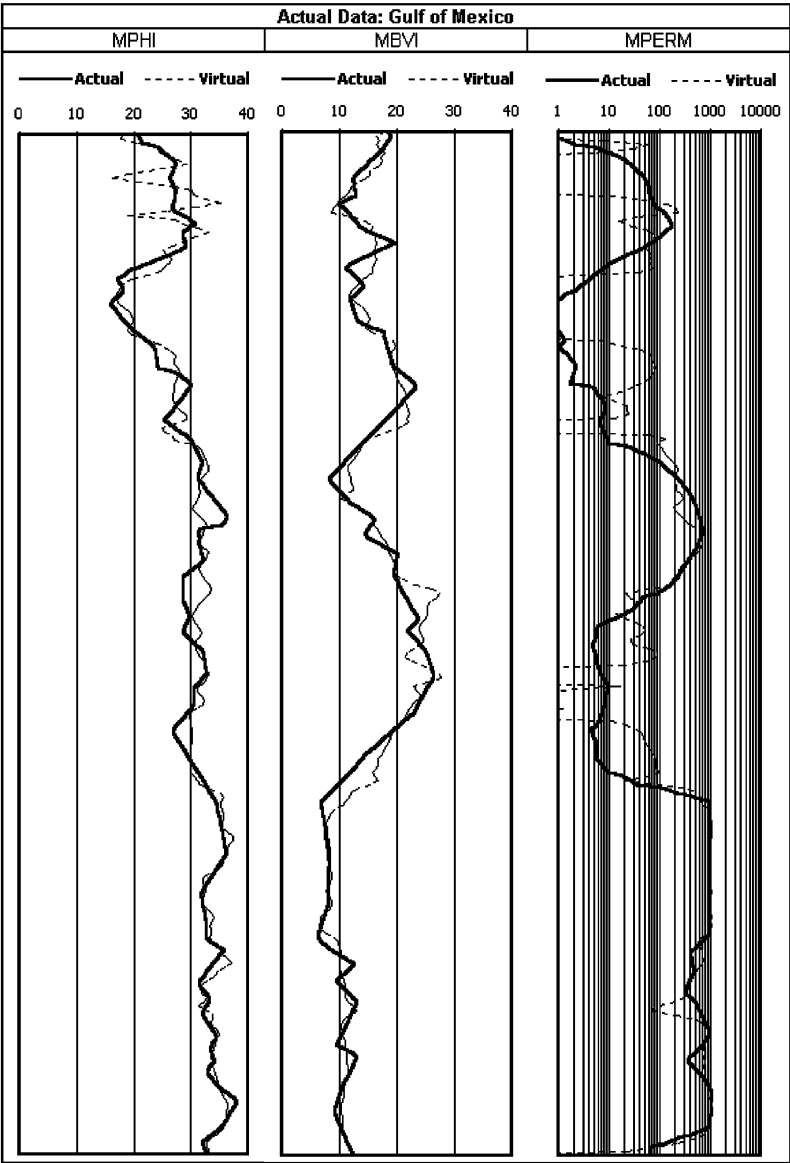


Fig. 5. Virtual and actual MRI logs for the entire pay zone for the well in Gulf of Mexico.

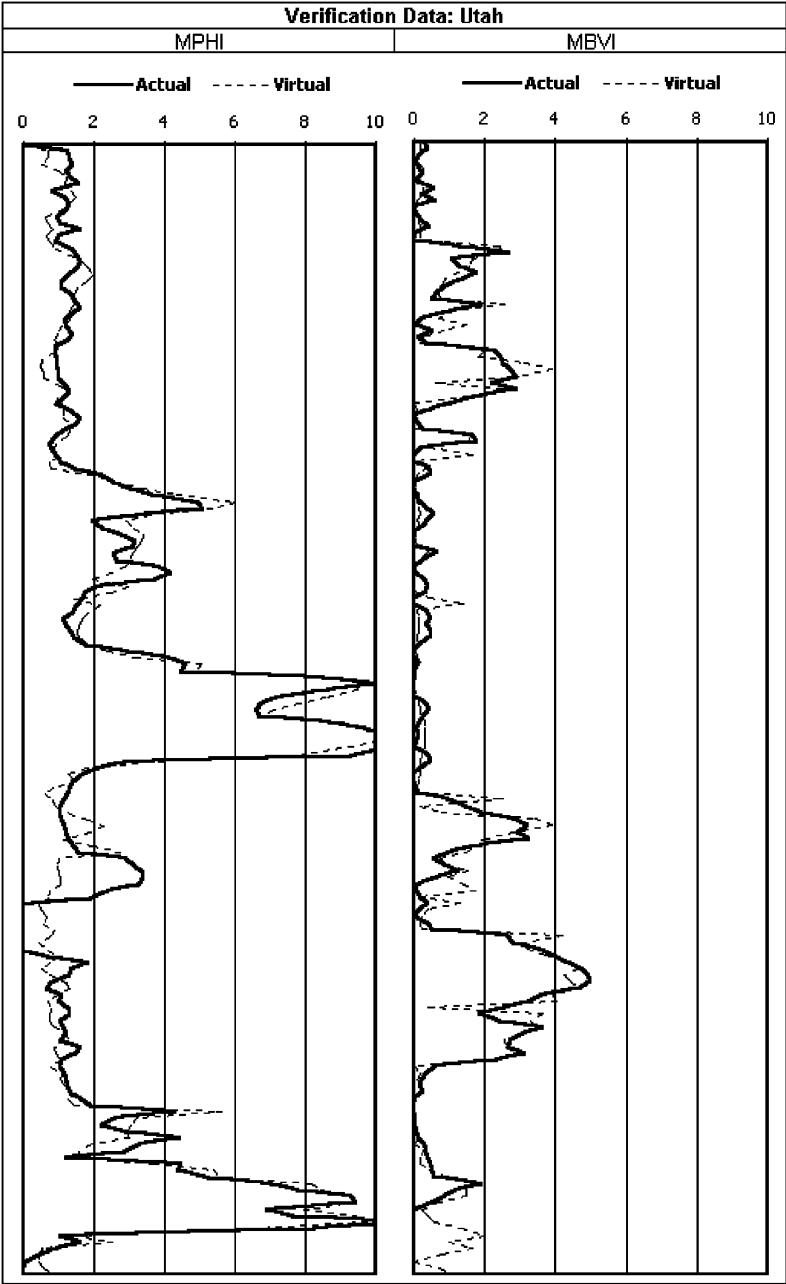


Fig. 6. Virtual and actual MRI logs for the verification data set for the well in Utah.

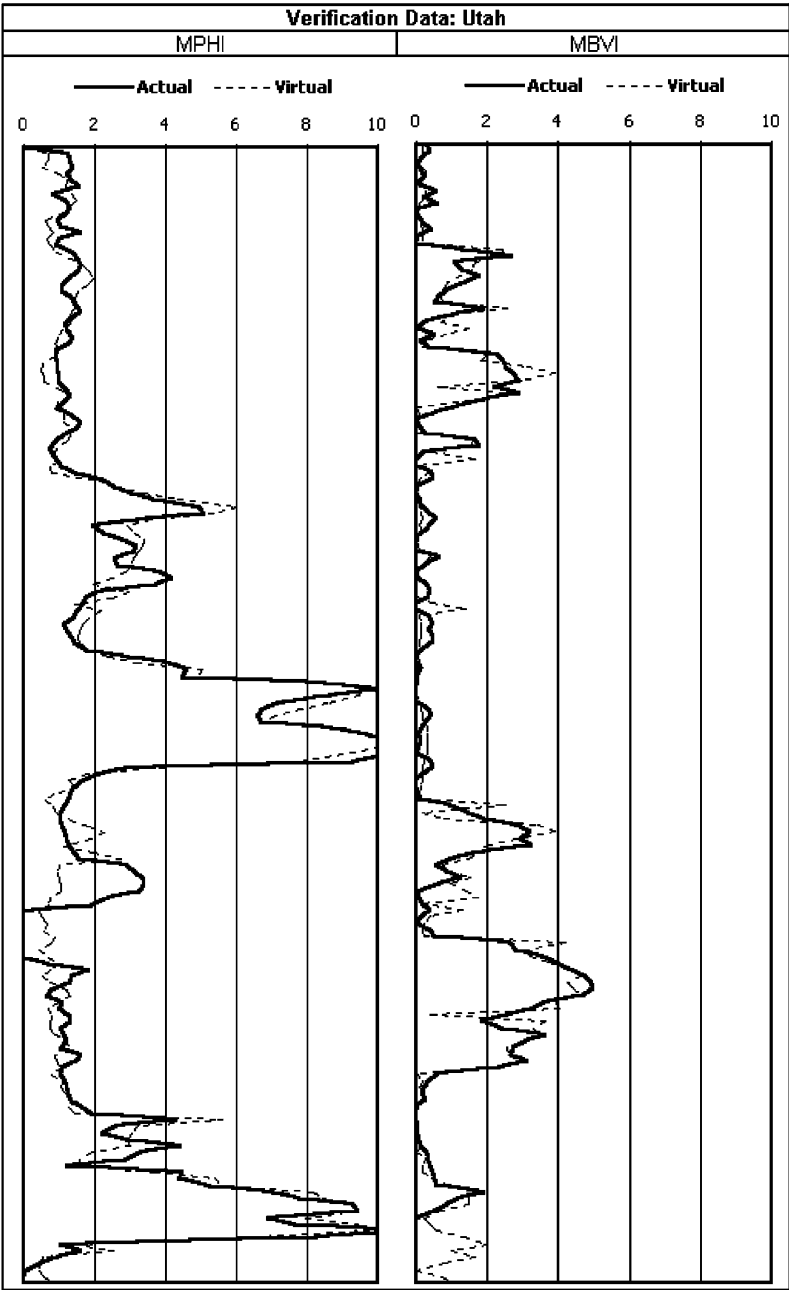


Fig. 7. Virtual and actual MRI logs for the entire pay zone for the well in Utah.

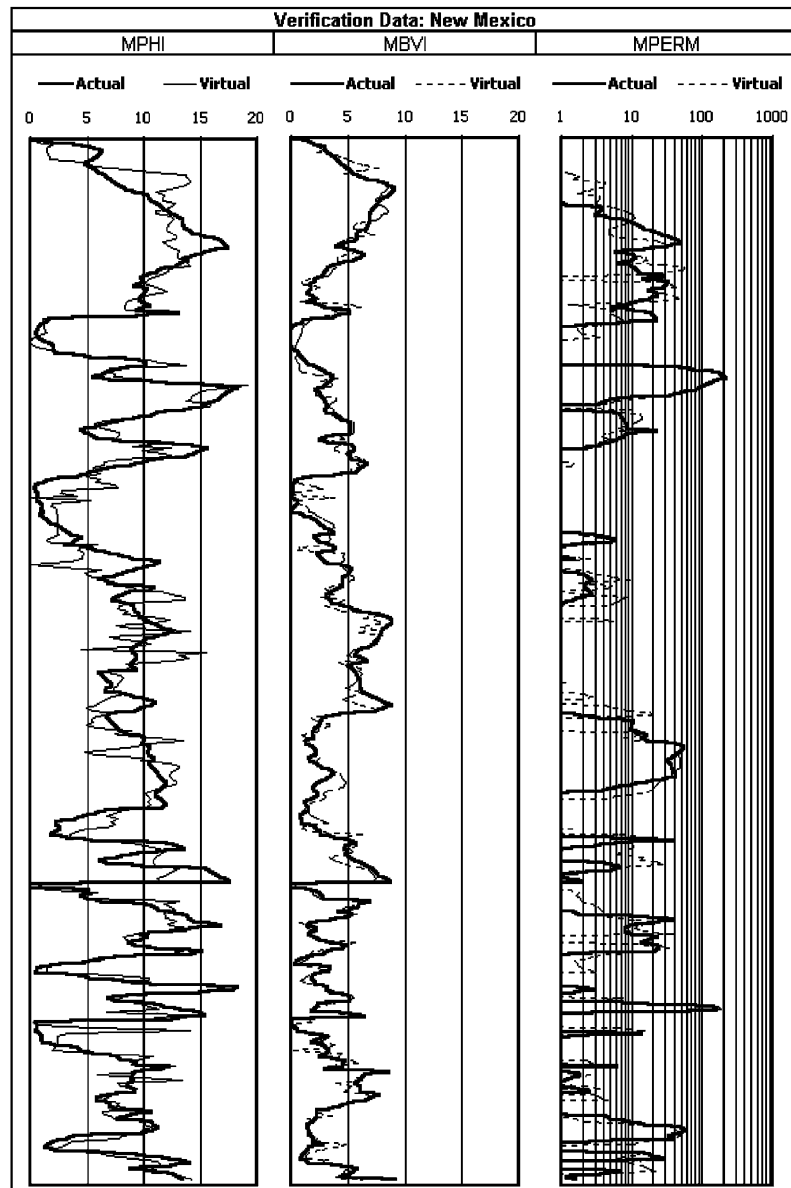


Fig. 8. Virtual and actual MRI logs for the verification data set for the well in New Mexico.

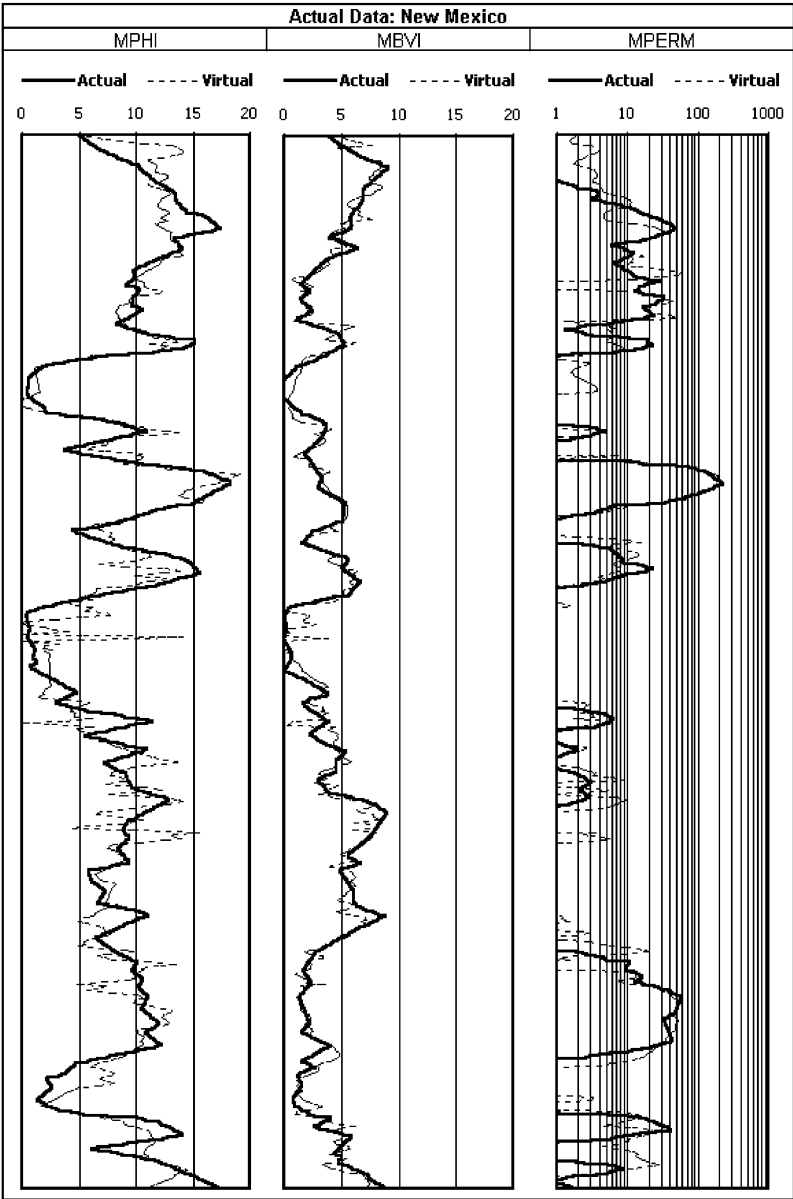


Fig. 9. Virtual and actual MRI logs for the entire pay zone for the well in New Mexico.

Fig. 2 shows only the verification data set—data never seen by the network before—while Fig. 3 contains the virtual and actual logs for the entire pay zone. Virtual effective porosity log (MPHI) has a correlation coefficient of 0.941 for the verification data set and a 0.967 correlation coefficient for the entire pay zone. The values for virtual MBVI log are 0.853 and 0.894, respectively. The virtual permeability log for this well also shows a strong correlation, 0.966 for the verification data set and 0.967 for the entire pay zone.

Figs. 4–9 show similar results for the wells from Utah, Gulf of Mexico, and New Mexico, respectively. In all cases shown in these figures, virtual MRI logs closely follow the general trends of the actual MRI logs. Please note that MPERM logs are shown in logarithmic scale and therefore the difference in the lower values of the permeability can be misleading. The correlation coefficient provides a more realistic mathematical measure of closeness of these curves to one another.

Table 1 is a summary of the analysis done on all four wells. This table contains the correlation coefficients for all the logs that were generated. This table

Table 1
Correlation coefficient between actual and virtual MRI logs

Well location	MRI log type	Data set	Correlation coefficient
Texas	MPHI	Verification	0.941
		Entire well	0.967
	MBVI	Verification	0.853
		Entire well	0.894
	MPERM	Verification	0.966
		Entire well	0.967
Utah	MPHI	Verification	0.800
		Entire well	0.831
	MBVI	Verification	0.887
		Entire well	0.914
	MPERM	Verification	0.952
		Entire well	0.963
Gulf of Mexico	MPHI	Verification	0.858
		Entire well	0.893
	MBVI	Verification	0.930
		Entire well	0.940
	MPERM	Verification	0.945
		Entire well	0.947
New Mexico	MPHI	Verification	0.957
		Entire well	0.960
	MBVI	Verification	0.884
		Entire well	0.926

Table 2

A per-acre estimate of the recoverable reserves using actual and virtual MRI logs

Well location	MRI type	Reserve Bbls/acre	Percent difference
Texas	Actual	52,368	– 1.4
	Virtual	51,529	
New Mexico	Actual	24,346	– 1.9
	Virtual	23,876	
Gulf of Mexico	Actual	240,616	+ 0.3
	Virtual	241,345	
Utah	Actual	172,295	– 1.8
	Virtual	169,194	

shows the accuracy of the virtual MRI log methodology on wells from different locations in the United States. The lowest correlation coefficient belongs to virtual MPHI log for the well located in Utah—0.800—while the best correlation coefficient belongs to virtual MPERM log for the well located in East Texas—0.966.

It should be noted that in this study absence of ideal data forced authors to use only 30% of the available data for construction and training of the neural networks. Authors believe that if more data were available from a particular field, the accuracy of the networks would increase.

Although the correlation coefficients for all the virtual logs are quite satisfactory, it should be noted that once these logs are used to calculate estimated recoverable reserves, the results are even more promising. This is due to the fact that many times the effective porosity and saturation is averaged. After all, MRI logs are used in two different ways. One way is to locate and complete portions of the pay zone that have been missed due to the conventional log analysis. This is more a qualitative analysis than a quantitative one since the engineer will look for an increase in the difference between MBVI and MPHI that correspond to a high permeability interval. The second use of these logs is to estimate the recoverable reserves more realistically.

The reserve estimates calculated using virtual MRI logs when compared to estimates calculated using actual MRI logs were quite accurate. As shown in Table 2, the reserve estimates using virtual MRI logs ranged from underestimating the recoverable reserves by 1.8% to over estimating it by 0.3%.

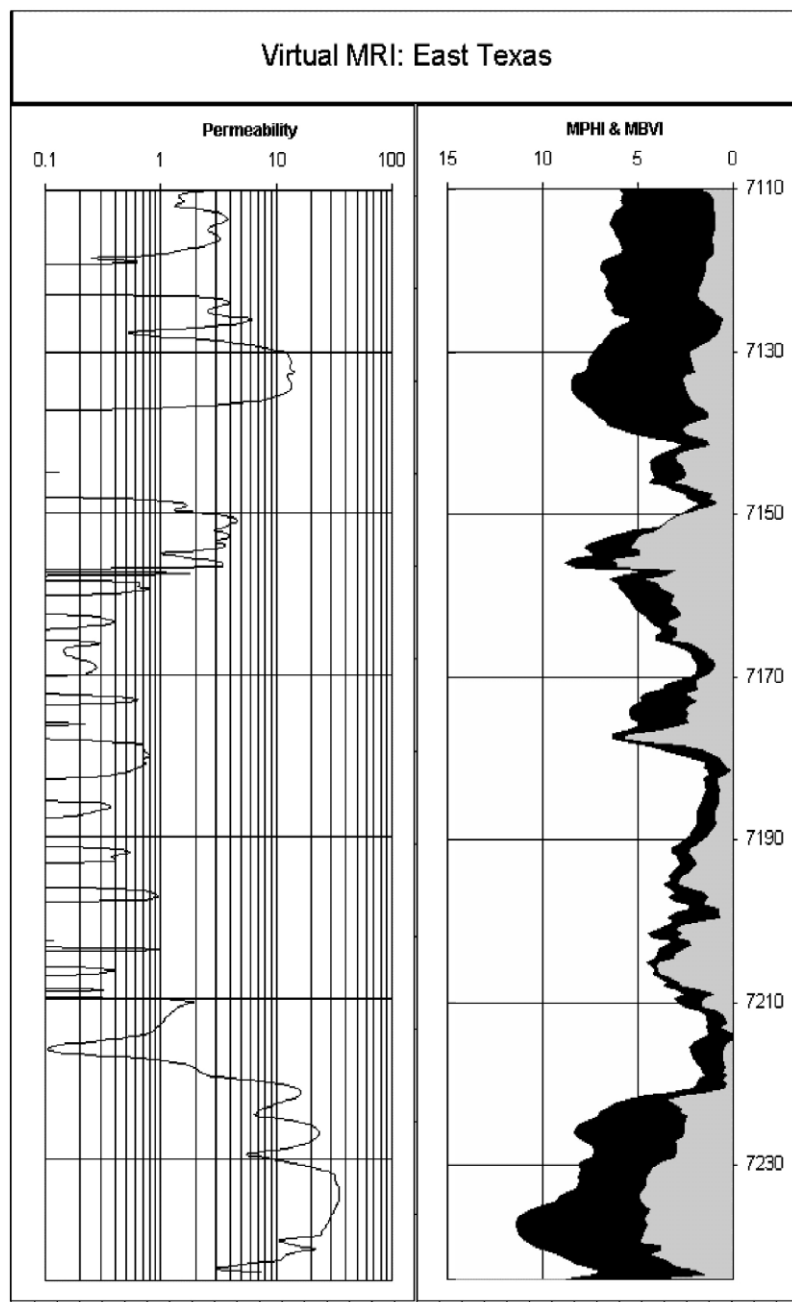


Fig. 10. Virtual MRI logs for the well in East Texas.

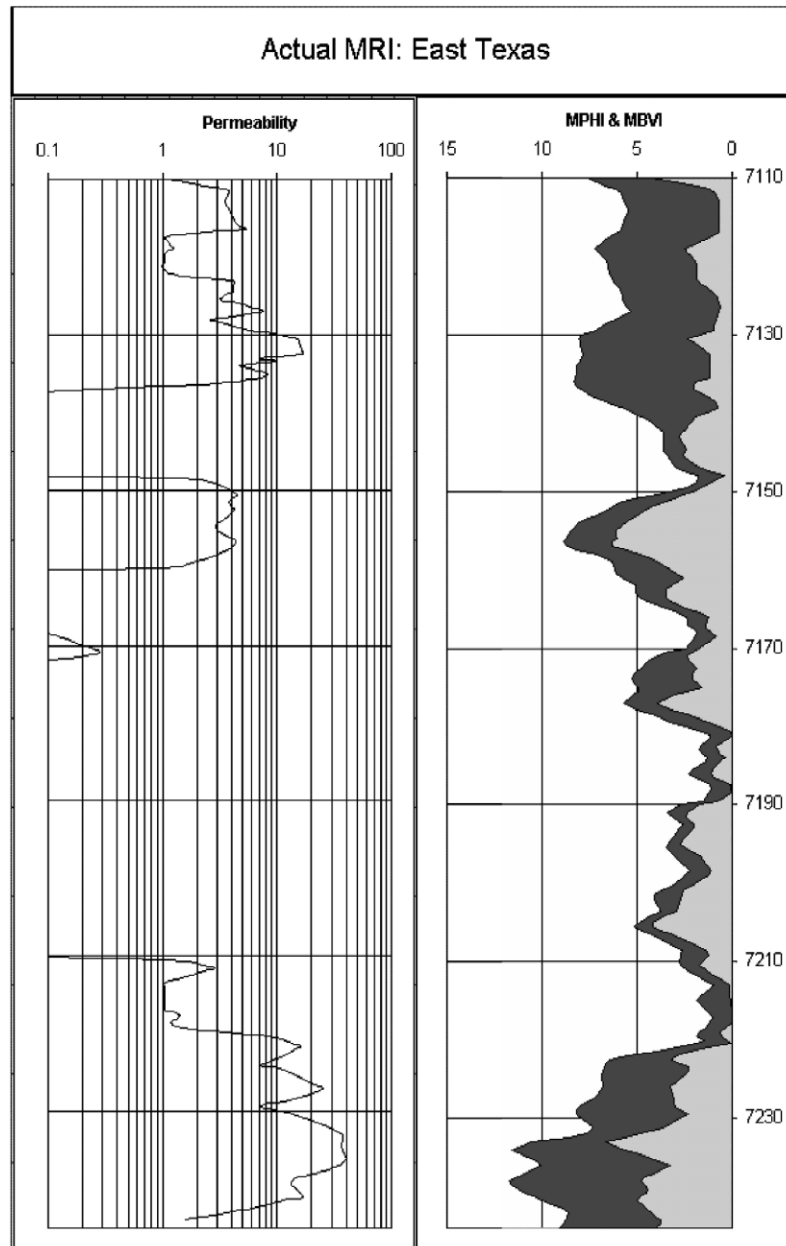


Fig. 11. Actual MRI logs for well in East Texas.

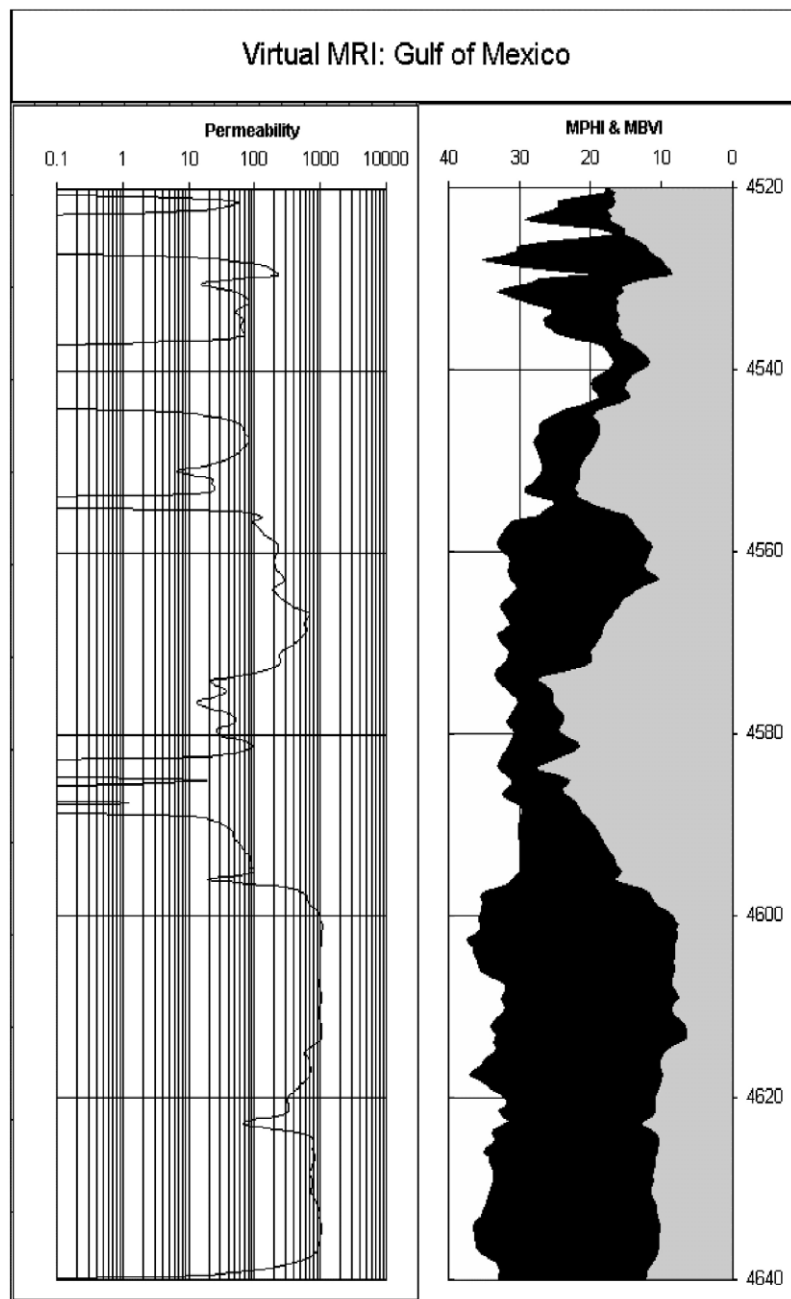


Fig. 12. Virtual MRI logs for the well in Gulf of Mexico.

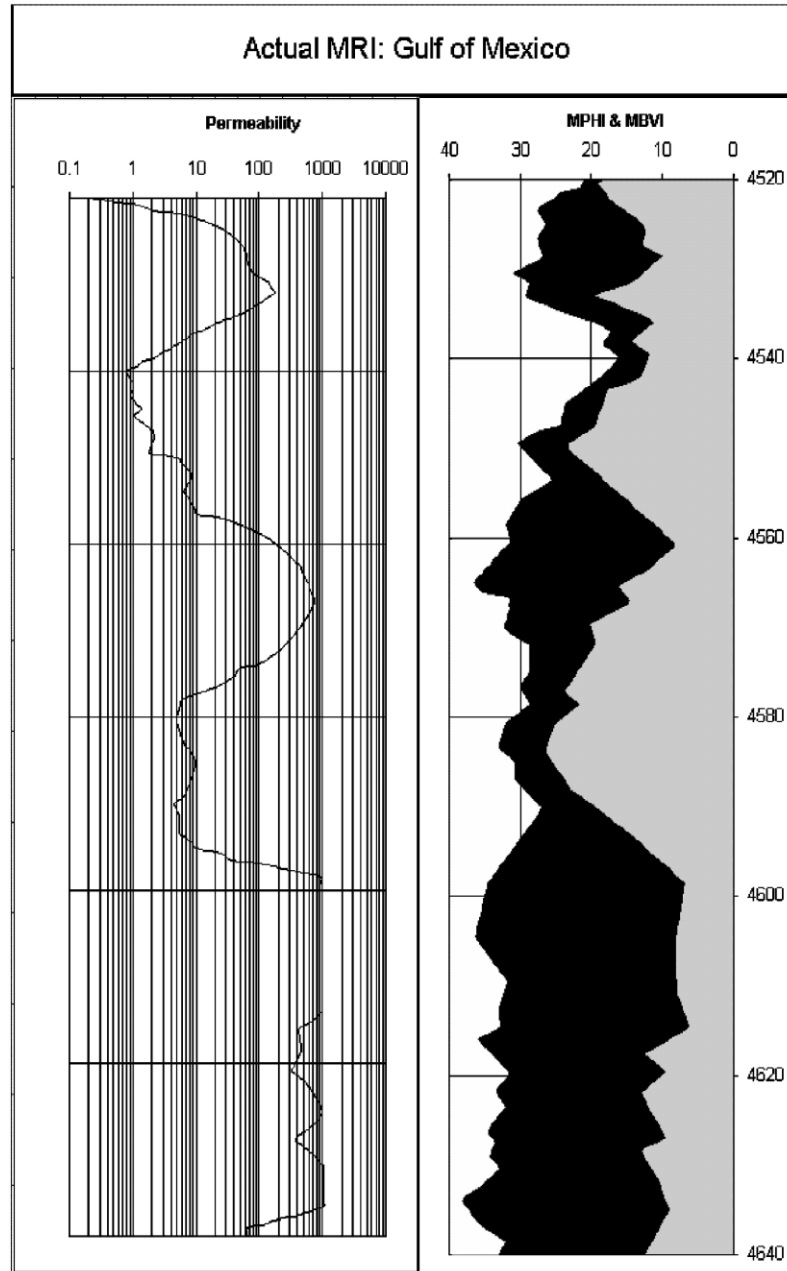


Fig. 13. Actual MRI logs for the well in Gulf of Mexico.

Figs. 10–13 show the virtual and actual MRI logs for wells in East Texas and the Gulf of Mexico. These logs are shown in the fashion that MRI logs are usually presented. These logs clearly show the free fluid index—difference between MBVI and MPHI logs—and the corresponding permeability values. This particular representation of the MRI logs is very useful to locate the portions of the pay zone that should be completed. The parts of the pay that has a high free fluid index and corresponds to a reasonably high permeability value are excellent candidates for completion.

5. Conclusions

In this paper, authors demonstrated that development of reasonably accurate virtual magnetic imaging logs is quite possible. The neural networks that are constructed and trained for a particular formation may not be used to generate virtual MRI logs for other formations. This conclusion is similar to the prior experience of the authors (Mohaghegh et al., 1996a). As in the case of virtual measurement of formation permeability, the methodology is formation-dependent.

Virtual MRI logs are valuable tools to accurately and realistically estimate recoverable reserves—especially in low resistivity formations—as well as identification of portions of the pay zone that should be completed. The most effective way of employing the virtual MRI logging method is to:

- Physically perform MRI logs on several strategically located wells in the field,

- Use the data to develop a virtual MRI tool for the formation, and
- Use the tool to generate virtual MRI logs for the remaining wells in the field.

Using virtual MRI logs methodology in field-wide bases can cut the cost considerably when compared to performing actual MRI logging on all the wells in a field. Authors also conclude—from our preliminary work that has not been shown in this paper—that similar methodology can be employed to generate many different kinds of virtual well logs such as SP, density porosity, etc., from other available logs.

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